Presentation of Technical Work

All reports submitted in the GE 213 Course should follow the Sandwich Presentation Method, which is appropriate to present a technical work. Thus the report must have the following items at least:

- Well formatted cover page
- Introduction
- Main body of the report
- Conclusions
- References

I. COVER PAGE

The cover page of the technical work should at least contain:

1. The title of the work: the title should be general (not the same as the problem words).
2. The names of the students who did the work, their computer numbers, their team number, and their section number.
3. The name of the course and its number.
4. Semester and year.
5. Date of submission.
6. The name of the course facilitator.
7. The name of the faculty and the university.

All the information above must be arranged with good page format. Some companies may have standard cover page and the writer may have to use it.

II. INTRODUCTION

The introduction orients the reader to the report (i.e. gives the reader some sense of what follows). This material could:

a) explain why the work is done (e.g., This work on frame weight is undertaken to get a better idea about how much this Home Exercise Machine might weigh), or

b) tell the reader what is accomplished (e.g., When this assignment is completed the student will have shown he knows how to do force balances and will be prepared for the quiz), or

c) tell the reader how the work fits into a bigger problem (e.g., Estimating the cost of the Home Exercise Machine requires knowing the weight of the machine's frame which is determined in the following work), or

d) tell the reader the important topics to follow and the order the topics are addressed (e.g., This report first discusses the methodology used, then presents the test results, and ends with a discussion of the testing)

III. MAIN BODY OF THE REPORT

1. Don't write "Body of work" as a title or a subtitle. Divide the main body into sections and use appropriate titles and subtitles.
2. Body part should contain detailed information.
3. Similar titles and similar subtitles must have similar format (i.e. if bold or underline is used in one subtitle then it must be used with all similar subtitles).
4. Try to use tables, charts, diagram, etc.., to make the subject understandable.
5. All tables and figures must be numbered. Do not put any of them unless you mention it in the text. They appear only after the paragraphs which refer to them.
6. Tables should have titles on the top, while figures have their titles at the bottom.
7. Use a level of language that the supposed reader can easily understand.
8. Main body may contain the following topics:
    - Problem definition and given data
    - Assumptions
    - Heuristics
    - Modeling steps and description
    - Equations
    - Calculations
    - Results and Discussions

IV. CONCLUSSIONS

This part ends the report that summaries and reflects the work done. You can use "Conclusion", , Conclusion and Recommendations, or "Summary" as a title or a subtitle.

This end material could reflects:

a) what was learned (e.g. The frame weight was 34 kilograms which is much lighter than any machine found in the market, or: This cone surface area of 250 m^2 is huge, about a third of a football field), or

b) the process used in working the problem (e.g. This problem could not be solved until it was realized that 1 should be replaced by $\sin 2\theta+\cos 2\theta$, or: Until the frame was simplified, no convergent model was obtained), or

c) the correctness of the result (e.g. The answer to this problem is 14.6 m^2, which matches the answer in the back of the book, or: This weight of 34 kilos, while it seems light, is probably correct; the model was checked using example problem 4.5 in Shigley), or

d) what will happen next (e.g. Now the cost of the device can be calculated).

V. REFERENCES

All references used must be listed at the end of the report. In the text the reference used are referred to by numbers put into square brackets, e.g. [5]. The list of reference should be in a standard form, for example:

1. Qassim University, Faculty of Engineering, Course: Introduction to Engineering Design II, First day materials, July 2010.
2. Korn, G. and Korn, T., Mathematical Handbook for Scientists and Engineers, 2nd Ed., McGraw-Hill, 1968.
3. Baumeister, E., Avallone, E., and Baumeister, T III, Marks, Standard Handbook for Mechanical Engineers, 8th Ed., McGraw-Hill, 1978.
4. Web site: http/www.???? , visited in 14/07/2010 at 10.45 PM.

VI. IMPORTANT POINTS TO BE NOTICED IN TECHNICAL WRITING

1. Some work may include, for instance, the proof of equation, data brought from external source or any complementary work . This is usually put in an Appendix after the whole report. An Appendix must start in a new page.
2. In your writing, one paragraph should contain only one subject. If the subject is changed, use another paragraph.
3. There should be logical relation between sentences in the paragraph.
4. Use punctuation as appropriate.
5. Do not use "I, we, or you". Use "it" as passive.
6. Write with suitable font size (12 pt. or 14 pt.) Times New Roman.
7. If you download anything from web pages do not use copy and paste. You must edit the text to be suitable for your report.

VII. EXAMPLE

An example for a technical work report will be given to you in the course web site. Your expected written technical work can be like this report, which just meets the expectation and gets M (Meet expectation). Any less than this work will result with NI (Need Improvement). More than this work may get E (Exceed expectation).

This presentation simply explains how to write a technical work. But, a good technical work should be better than this explanation. The main points of the written technical work are given. A specific technical work may require some other rules.